

Programming Fundamentals

Assignment Of Lists

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Program#1: Given a=[1,2,4,2,5,2,6], remove all 2s and print the final list.

```
1 a = [1, 2, 4, 2, 5, 2, 6]
2 while 2 in a:
3     a.remove(2)
4 print(a)
```

Output

```
[1, 4, 5, 6]
```

```
=== Code Execution Successful ===
```

Program#2: Given nums=[9,2,11,7,11,3], find the second largest distinct value using loops

```
1 second=0
2 nums=[9,2,11,7,11,3]
3 largest=0
4 for x in nums:
5     if x>largest:
6         second=largest
7         largest=x
8     elif largest>x>second:
9         second=x
10 print(second)
```

Output

9

=== Code Execution Successful ===

Program#3: Given `a=[10,20,30,40,50]`, rotate left by 1: result should be `[20,30,40,50,10]`

```
1 a = [10, 20, 30, 40, 50]
2 first = a.pop(0)
3 a.append(first)
4 print(a)
```

Output

`[20, 30, 40, 50, 10]`

=== Code Execution Successful ===

Program#4: Given `a=[1,2,3,4,5]`, insert value 99 at index k (input k). Do not use `insert()`.

```
1 a = [1, 2, 3, 4, 5]
2 k = int(input("Enter index k: "))
3 value = 99
4 a = a[:k] + [value] + a[k:]
5 print(a)
```

Output

Enter index k: 3

`[1, 2, 3, 99, 4, 5]`

=== Code Execution Successful ===

Program#5: Given a=[5,6,7,8,9,10], delete the element at index k using shifting + final pop()

```
main.py
1 a = [5, 6, 7, 8, 9, 10]
2 k = int(input("Enter index k: "))
3 del a[k]
4 print(a)
```

```
Output
Enter index k: 2
[5, 6, 8, 9, 10]

=== Code Execution Successful ===
```

Program#6: Input a line of text. Split into words. Print the longest word and its length

```
1 line = input("Enter a line: ")
2 words = line.split()
3 longest = max(words, key=len)
4 print("Longest:", longest)
5 print("Length:", len(longest))
```

```
Output
Enter a line: My name is Laiba
Longest: Laiba
Length: 5

=== Code Execution Successful ===
```

Program#7: Given a sentence, split into words, then join using " | " as separator and print

```

1 s = input("Enter sentence: ")
2 result = " | ".join(s.split())
3 print(result)

```

Output

```

Enter sentence: My name is Laiba
My | name | is | Laiba

```

=== Code Execution Successful ===

Program#8: Given a=[3,3,2,3,1,2,4,4], create a new list of unique elements preserving order

```

1 a = [3, 3, 2, 3, 1, 2, 4, 4]
2 u = []
3 for x in a:
4     if x not in u:
5         u.append(x)
6 print(u)

```

Output

```
[3, 2, 1, 4]
```

=== Code Execution Successful ===

Program#9: Given a=[1,1,1,2,2,3,4,4], print each distinct value with its frequency

```

1 a = [1, 1, 1, 2, 2, 3, 4, 4]
2 for x in set(a):
3     print(x, "->", a.count(x))

```

Output

```
1 -> 3
2 -> 2
3 -> 1
4 -> 2
```

=== Code Execution Successful ===

Program#10: Sort a=[8,3,5,2,9,1] in ascending order using bubble sort (no sort())

```
1 a = [8, 3, 5, 2, 9, 1]
2 n = len(a)
3 for k in range(n):
4     for i in range(0, n - k - 1):
5         if a[i] > a[i+1]:
6             a[i], a[i+1] = a[i+1], a[i]
7 print(a)
```

Output

```
[1, 2, 3, 5, 8, 9]
```

=== Code Execution Successful ===